

PRACTICE WORKSHEET – INFORMATICS PRACTICES (IP)

CLASS XII

Topic 1: Database Concepts & Relational Data Model

1. State whether the following statement is **True or **False**:**

A table can have more than one primary key.

2. Which of the following represents a **relation in a relational database?**

- a) A row in a table
 - b) A column in a table
 - c) A table with rows and columns
 - d) A database name
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3. What is the full form of **DML in SQL?**

- a) Data Management Language
 - b) Data Manipulation Language
 - c) Database Modification Language
 - d) Domain Manipulation Language
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4. Which SQL command is used to **remove all records from a table but keep the table structure?**

- a) DELETE
 - b) DROP
 - c) REMOVE
 - d) TRUNCATE
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5. Which of the following is **not a valid SQL numeric data type?**

- a) INT
 - b) DECIMAL
 - c) CHAR
 - d) FLOAT
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6. The **cardinality of a relation refers to:**

- a) The number of columns in a table
 - b) The number of rows in a table
 - c) The primary key of the table
 - d) The data type of attributes
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7. State whether the following statement is **True or **False**:**

SELECT * FROM Products; will display only the first column of the Products table.

8. Which SQL clause is used to **sort the result** of a query?

- a) GROUP BY
 - b) ORDER BY
 - c) WHERE
 - d) HAVING
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9. Which of the following is a **valid relational operator** used in SQL WHERE clause?

- a) >=
 - b) = =
 - c) :=
 - d) != =
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10. Which keyword is used to **display only unique values** from a column?

- a) UNIQUE
 - b) DIFFERENT
 - c) DISTINCT
 - d) ONLY
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11. Which of the following commands is used to **remove a column** from an existing table?

- a) DELETE COLUMN
 - b) ALTER TABLE ... DROP COLUMN
 - c) REMOVE COLUMN
 - d) DROP COLUMN
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12. Which of the following is **not** a valid SQL keyword?

- a) BETWEEN
 - b) IN
 - c) LIKE
 - d) FOR
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13. What does the **IN** operator do in SQL?

- a) Checks if a value is NULL
 - b) Checks if a value matches any value in a list
 - c) Checks if two values are equal
 - d) Checks if a value is within a range
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14. Assertion (A): Every table must have a primary key.

Reason (R): A primary key uniquely identifies each row in a table.

Choose the correct option:

- a) Both A and R are True and R is the correct explanation for A

- b) Both A and R are True but R is not the correct explanation for A
- c) A is True but R is False
- d) A is False but R is True
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15. Which SQL command is used to **delete an entire database**?

- a) DELETE DATABASE
- b) DROP DATABASE
- c) REMOVE DATABASE
- d) ERASE DATABASE
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16. State whether the following statement is **True** or **False**:

In a relational database, each column in a table is called an attribute.

17. Which of the following is a **valid SQL date and time data type**?

- a) INT
- b) VARCHAR
- c) DATE
- d) TEXT
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Topic 2: DDL Statements (CREATE, ALTER, DROP)

18. Write SQL statements for the following tasks:

I. Create a database named **UniversityDB**.

II. Create a table **Teacher** with the following structure:

Column Name	Data Type
TeacherID	INT
TeacherName	VARCHAR(35)
Subject	VARCHAR(25)
Salary	DECIMAL(10,2)

19. Write SQL statements for the following tasks:

I. Add a new column **Experience** of type INT to the Teacher table.

II. Change the data type of **Salary** column to FLOAT.

Topic 3: DQL Statements (SELECT, FROM, WHERE)

20. Consider the following table **Employee**:

EmpCode	EmpName	City	Salary
201	Ankit	Pune	55000
202	Sneha	Mumbai	72000
203	Raj	Pune	48000

204	Neha	Delhi	65000
205	Amit	Mumbai	70000

Write SQL queries for the following:

- I. Display the names of all employees living in **Mumbai**.
- II. Display the **EmpCode** and **EmpName** of employees whose salary is greater than 60000.
- III. Display all employees whose salary is between 50000 and 70000.

21. Consider the following table **Products**:

ProductCode	ProductName	Category	Price
P01	Laptop	Electronics	65000
P02	Mouse	Electronics	600
P03	Chair	Furniture	3500
P04	Keyboard	Electronics	1500
P05	Table	Furniture	8000

Write SQL queries for the following:

- I. Display all records where **Category** is "Electronics".
- II. Display **ProductName** and **Price** for products with Price greater than 1000.
- III. Display all products where **Price** is less than 5000.

Topic 4: Candidate Key, Primary Key, Alternate Key

- 22. A.** What is a **Primary Key**? Explain with an example.
- B.** Differentiate between **Candidate Key** and **Primary Key**.

Topic 5: Output Prediction (SELECT with WHERE and BETWEEN)

23. Consider the following table **Book**:

BookID	Title	Author	Pages
101	Python Guide	Meera	450
102	SQL Basics	Raj	300
103	Java Master	Amit	600
104	Web Design	Neha	400
105	AI Concepts	Raj	550

Write the output of the following SQL queries:

- I. SELECT * FROM Book WHERE Pages BETWEEN 350 AND 500;
- II. SELECT Title FROM Book WHERE Author = "Raj";
- III. SELECT * FROM Book ORDER BY Pages DESC;

Topic 6: Relational Data Model Terms

24. A. Explain the following terms with suitable examples:

I. Tuple

II. Domain

III. Degree of a relation

B. What is a **Candidate Key**? How is it different from an **Alternate Key**? Explain with an example.

Topic 7: SQL Queries with Logical Operators

25. Consider the following table **Student**:

RollNo	Name	Class	City	Marks
1	Kavya	11	Delhi	88
2	Rohan	12	Mumbai	76
3	Priya	12	Delhi	92
4	Ankit	11	Pune	81
5	Sneha	12	Mumbai	79

Write SQL queries for the following:

I. Display all students whose **Class** is 12 and **Marks** is greater than 80.

II. Display **Name** and **Marks** of students who live in **Delhi** or **Mumbai**.

III. Display all students whose **Marks** is between 75 and 85.

IV. Display all students sorted by **Name** in ascending order.

Topic 8: ALTER TABLE Operations

26. Write SQL statements to perform the following operations:

I. Create a database named **LibraryDB**.

II. Create a table **Member** with the following structure:

Column Name	Data Type
MemberID	INT
MemberName	VARCHAR(30)
MembershipType	VARCHAR(20)
Fees	DECIMAL(8,2)

III. Add a new column **PhoneNo** of type VARCHAR(15) to the Member table.

IV. Change the data type of **Fees** column to INT.

Topic 9: Output Prediction (DISTINCT, IN, AND, OR)

27. Predict the output of the following SQL queries based on the **Employee** table shown below:

Table: Employee

EmpID	EmpName	Department	Salary
101	Aarav	IT	85000
102	Meera	HR	72000

103	Raj	IT	78000
104	Neha	Finance	69000
105	Amit	HR	75000

Write the output for each query:

- I. SELECT DISTINCT Department FROM Employee;
- II. SELECT EmpName, Salary FROM Employee WHERE Salary > 75000 AND Department = "IT";
- III. SELECT * FROM Employee WHERE Department IN ("HR", "Finance");
- IV. SELECT * FROM Employee WHERE Department = "IT" OR Salary > 80000;

Topic 10: SQL Queries with ORDER BY

28. Consider the following table **Sales**:

SaleID	ItemName	Category	Qty	UnitPrice
201	Laptop	Electronics	4	55000
202	Mouse	Electronics	15	400
203	Pen	Stationery	60	25
204	Keyboard	Electronics	5	1300
205	Notebook	Stationery	40	35

Write SQL queries for the following:

- I. Display all records where **Category** is "Stationery".
- II. Display **ItemName** and **Qty** for items with Qty greater than 8.
- III. Display all items where **UnitPrice** is less than 2000.
- IV. Display all items sorted by **ItemName** in descending order.

Topic 11: SQL Queries (String Functions, Date)

29. A. Write suitable SQL queries for the following tasks:

- I. To display the first four characters of the **ItemName** column from the **Inventory** table.
- II. To display the **CustomerName** column from the **Customers** table.
- III. To display the **Address** column.
- IV. To display all columns from the **Transactions** table where the **TransactionDate** is after '2024-01-01'.

B. Consider the table **Staff** with columns: **StaffID**, **StaffName**, **Department**, **Salary**, **JoiningDate**.

Write SQL queries for the following:

- I. Display all staff whose **Department** is "Sales" and **Salary** is greater than 50000.
- II. Display the **StaffName** and **JoiningDate** of staff who joined after **2022-06-01**.
- III. Display all staff whose **Department** is either "Marketing" or "IT".
- IV. Display all staff sorted by **StaffName** in ascending order.
- V. Display all staff whose **StaffName** starts with the letter 'S'.